

Assignment: Unsupervised Learning Integrated Mini Project

Objective

Develop **ONE real-world application** that integrates the following **four types of Unsupervised Learning**:

1. Clustering
2. Dimensionality Reduction
3. Anomaly Detection
4. Association Rule Learning

Your project should demonstrate how these techniques can be combined to solve a **real-world problem** using **Python and Gradio**.

Project Requirements

Your project must:

- Apply **Clustering** to identify groups or patterns in the data.
- Apply **Dimensionality Reduction** to reduce the number of features or visualize high-dimensional data.
- Apply **Anomaly Detection** to identify unusual or abnormal data points.
- Apply **Association Rule Learning** to discover relationships or patterns between items/features.
- Include a **Gradio user interface (UI)**.
- Provide meaningful outputs/results for the selected real-world application.
- Be implemented using **Python**.
- Run successfully in **Google Colab or VS Code**.

Suggested Project Areas

You may choose a real-world application such as:

- Customer Analytics
- Retail / E-commerce
- Banking / Finance

Assignment: Unsupervised Learning Integrated Mini Project

- Healthcare
- Education
- E-commerce Recommendation
- Cybersecurity
- Social Media Analytics

You are also encouraged to propose your **own real-world application**.

Submission Requirements

Submit **ONE Python file (.py)** containing your complete project.

The Python file should include:

- Data loading and preprocessing
- Clustering implementation
- Dimensionality Reduction implementation
- Anomaly Detection implementation
- Association Rule Learning implementation
- Gradio interface
- Complete executable code

Suggested Filename

StudentName_UnsupervisedLearning_Project.py

Submission

- Source Code (.ipynb/.py)
- Report (1 Page)
- Screenshots of results

Submit your files before the deadline.

Please make sure that your project runs without errors and that all four Unsupervised Learning techniques are included in the project.